
Is the Reconstruction–Forecasting Tradeoff a Frontier or an Operating Point?

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Abstract

1 Autoencoder-based temporal models are routinely trained on a weighted sum of
2 a reconstruction loss and a forecasting loss, with the weight treated as a hyperpa-
3 rameter. This practice assumes the balance is a genuine frontier. We ask whether
4 it is instead a suboptimal operating point induced by forcing one latent to serve
5 both objectives. On a controlled chaotic benchmark (Lorenz-63 lifted to 30 noisy
6 channels, known intrinsic dimension 3) we split the representation into an un-
7 bounded 3-d carrier read only by the forecaster and a sigmoid-bounded 8×8
8 carrier read only by the decoder, trained in two stages so that the forecast gradient
9 provably cannot reach the decoder. At matched parameters, matched wall clock and
10 matched seeds, the split Pareto-dominates *every* setting of the weighted-loss knob
11 that forecasts at all. We then report the result that complicates this: the advantage
12 is compute-dependent. Given $10\times$ the wall clock, the balancing baseline overtakes
13 the split on both headline axes. What does not transfer with compute is stability
14 — bounded decoding keeps out-of-range rollouts at 0% against 100% for every
15 unbounded baseline. We conclude the tradeoff is not one fixed frontier, and argue
16 that compute-matched comparison alone cannot settle whether it is dissolvable.

17 1 Introduction

18 A large body of work trains one encoder to serve two objectives at once: reconstruct the present and
19 forecast the future. The standard mechanism is a weighted loss, $\mathcal{L} = (1 - \phi)\mathcal{L}_{\text{rec}} + \phi\mathcal{L}_{\text{fctst}}$, with ϕ
20 tuned per dataset. Coupled Attention Networks [Coupled Attention, 2023] state the weights explicitly
21 as “a tradeoff between the prediction and reconstruction models”; CAAE [CAAE, 2026] names the
22 underlying “objective mismatch between training for reconstruction fidelity and [the downstream
23 task]”; a broad cluster of reconstruction-plus-prediction hybrids follows the same recipe [MemAAE,
24 2021, HyVAE, 2023]. In physics-informed sequence modelling, SSP [Anonymous, 2026] pursues
25 the same separation by a different route — spectral truncation — describing its contribution as
26 “decoupling reconstruction fidelity from rollout regularity.”

27 Across these paradigms the tradeoff is treated as something to *tune*. Almost nobody asks whether the
28 tuning is necessary. That is the question here.

29 **What we are not claiming.** Rate–distortion theory [Shannon, 1959, Berger, 1971] proves a frontier
30 between rate and distortion for a given source, and the information bottleneck [Tishby et al., 2000]
31 extends this to predictive representations. We are not claiming to beat any such bound. Rate–distortion
32 constrains distortion *at a fixed rate*; it says nothing about which point inside the achievable region a
33 given architecture and schedule lands on. Our claim is only about the latter: at fixed capacity and

fixed compute, a shared latent under a weighted loss does not reach the best available point, and separating the carriers moves it.

Contributions. (i) We give a controlled, compute-matched test of whether the reconstruction–forecasting tradeoff is a frontier or an operating point, on a system whose true intrinsic dimension is known. (ii) We show a bounded/unbounded latent split, trained in two stages, Pareto-dominates every weighted-loss setting that forecasts at all, at matched parameters, wall clock and seeds. (iii) We report a negative result that constrains this: the dominance does not survive a $10\times$ compute budget for the baseline. (iv) We isolate which architectural ingredient buys what, using a bounded-shared-latent control, and find boundedness and separation buy different things.

2 Method

Objects. An observation is $x_t \in \mathbb{R}^n$. Five learned maps plus a forecaster:

$$\begin{aligned} E : \mathbb{R}^n &\rightarrow \mathbb{R}^n, & f : \mathbb{R}^n &\rightarrow \mathbb{R}^k, & r : \mathbb{R}^n &\rightarrow [0, 1]^{a \times b}, \\ m : \mathbb{R}^k &\rightarrow [0, 1]^{a \times b}, & g : \mathbb{R}^k &\rightarrow \mathbb{R}^k, & D : [0, 1]^{a \times b} &\rightarrow \mathbb{R}^n. \end{aligned}$$

Write $h_t = E(x_t)$, $b_t = f(h_t)$, $C_t = m(b_t)$. Here b is the *unbounded* carrier ($k = 3$) and C the *bounded* carrier ($a = b = 8$). The reconstruction path is $x_t \rightarrow h_t \rightarrow b_t \rightarrow C_t \rightarrow \hat{x}_t$; the forecast path steps only b and recomputes C from it, $b_{t+1} = g(b_t)$, $C_{t+1} = m(b_{t+1})$, $\hat{x}_{t+1} = D(C_{t+1})$.

Consistency. The chain is only coherent if the indirect route to the bounded latent agrees with the direct one, $m(f(h_t)) \approx r(h_t)$. Since r sees all n dimensions and m sees only k , enforcing agreement forces f to retain everything C needs while compressing $30 \rightarrow 3$. r is a training-only teacher and never runs at inference.

Two stages, and why the split is exact. Stage 1 trains $\theta_A = \{E, f, m, r, D\}$ on single timesteps: $\mathcal{L}_A = \|D(m(f(E(x_t)))) - x_t\|^2 + \lambda \|m(f(E(x_t))) - r(E(x_t))\|_F^2$. Stage 2 *freezes* θ_A , runs the frozen encoder over every training timestamp to obtain the series b_t , and trains g alone, free-running, entirely in b -space:

$$\mathcal{L}_B(\theta_g) = \frac{1}{H} \sum_{\tau=1}^H \|\hat{b}_{t+\tau} - b_{t+\tau}\|^2, \quad \hat{b}_{t+\tau} = g^{\circ \tau}(b_t). \quad (1)$$

Neither m nor D appears in $\mathcal{L}_B$, so $\partial \mathcal{L}_B / \partial \theta_A \equiv 0$: improving the forecast *cannot* degrade reconstruction. This is the decoupling, and it is an identity rather than a penalty. Contrast the weighted loss, where $\partial \mathcal{L}_{\text{fcst}} / \partial \theta_{\text{enc}} \neq 0$ and ϕ merely chooses which objective loses.

Boundedness gives unconditional stability. m ends in a sigmoid, so $C_{t+\tau} \in [0, 1]^{a \times b}$ by construction for any input. D is continuous on that compact set, hence $\|\hat{x}_{t+\tau}\| \leq \sup_{C \in [0, 1]^{a \times b}} \|D(C)\| < \infty$ for all τ , *independently of* $\hat{b}_{t+\tau}$. However far the unbounded carrier drifts, the decoded trajectory cannot leave a bounded set: error saturates instead of exploding. This costs zero parameters.

3 Experimental setup

Data. Lorenz-63 [Lorenz, 1963] at $dt = 0.01$, lifted through a frozen random tanh map to $n = 30$ channels and corrupted with Gaussian noise at 5% of each channel’s spread. True intrinsic dimension is 3, so k is a probe rather than a hyperparameter. Training pools 8 independent trajectories, each split 70/15/15 chronologically; 32 further trajectories are held out entirely, never split and never seen in training. Every standardisation constant is fit on the pooled training portion only. The reconstruction floor is *measured* from the realised noise draw in the final standardised scale (0.002492) rather than assumed as noise².

Table 1: Matched parameters, matched wall clock, matched seeds where marked. $\times$ floor is reconstruction MSE over the measured noise floor (lower better); VPT in Lyapunov times (higher better); divergence is the fraction of rollouts leaving the training range. The $10\times$ budget row is the *off-protocol* control (see “The negative result), not a matched comparison.

Model	seeds	$\times$ floor $\downarrow$	VPT $\uparrow$	NRMSE@5LT $\downarrow$	diverge $\downarrow$
Ours (two-stage)	3	1.06	0.489 $\pm$ 0.013	1.18	0%
$\phi=0$ (no forecasting)	1	1.01	0.027	30.6	100%
$\phi=0.1$	3	1.08	0.323 $\pm$ 0.011	15.5	100%
$\phi=0.25$	3	1.15	0.320 $\pm$ 0.015	14.9	100%
$\phi=0.5$	3	1.32	0.326 $\pm$ 0.013	15.8	100%
$\phi=0.9$	1	2.01	0.335	11.1	100%
$\phi=1$	1	3.79	0.326	13.4	100%
AEGRU+sigmoid (D)	3	2.28	0.329 $\pm$ 0.004	1.28	1.6%
PINN (true physics)	3	1.87	0.378 $\pm$ 0.019	1.19	26%
PINN ($\rho=26$)	1	2.50	0.245	1.27	0%
persistence / climatology	–	–	0.036 / 0.000	1.37 / 1.00	0%
$\phi=0.25$, $10\times$ budget	1	0.99	0.526	1.43	22%

Baselines. (B) the weighted-loss AE+GRU, ϕ swept over $\{0, 0.1, 0.25, 0.5, 0.75, 0.9, 1\}$, 16-d shared latent; (D) the same construction with its *shared* latent sigmoid-bounded and kept bounded through rollout by a residual step in logit space — the control that holds boundedness fixed and removes the split; (C) a physics-informed AE whose 3-d latent is calibrated to true Lorenz coordinates and rolled out by RK4 on the known equations, with a misspecified $\rho = 26$ variant; plus persistence and climatology.

Protocol. Every model gets **124,482 trainable parameters and 120 s of wall clock on one pinned CPU thread**. Ours, the PINN, $\phi \in \{0.1, 0.25, 0.5\}$ and model D run 3 seeds; remaining ϕ settings run 1. We report reconstruction MSE relative to the measured floor; valid prediction time (VPT, the lead time at which normalised error first crosses 0.4, in Lyapunov times [Vlachas et al., 2020]); normalised error at 5 Lyapunov times; and *divergence*, the fraction of rollouts leaving the training bounding box.

4 Results

At matched compute, the split dominates. Table 1 shows the split reaching $1.06\times$ the floor at VPT 0.489. Every ϕ setting that forecasts at all is worse on *both* axes simultaneously: the best of them reaches VPT 0.335, and the only setting with better reconstruction ($\phi=0$, $1.01\times$) has VPT 0.027, below persistence — it does not forecast. So the split is not a different point on the knob’s curve; it is outside the region the knob reaches. The knob’s own curve is also nearly flat in VPT (0.320–0.335 across $\phi \in [0.1, 1]$) while reconstruction degrades by $3.8\times$, i.e. within this family, paying reconstruction buys almost no forecast.

The negative result: this is compute-dependent. Held at 120 s, the above reads as dissolution. It is not that simple. Given $10\times$ the wall clock, $\phi=0.25$ reaches VPT 0.526 at $0.99\times$ the floor — better than the split on both headline axes. The shift ($\Delta\text{VPT} = 0.205$) is an order of magnitude larger than the 3-seed spread of the $1\times$ run ($\sigma = 0.015$), so the weighted-loss family is *measurably undertrained* at the matched budget, not frontier-limited there. We report this prominently because it bounds the claim: a compute-matched Pareto comparison, the standard protocol in this literature, can invert under a compute rescaling. Any claim of the form “architecture X dissolves the tradeoff,” ours included, is only as strong as the budget it was measured at.

What does not follow compute: stability. Divergence separates the models in a way compute does not fix. Every unbounded baseline leaves the training range on 100% of rollouts at $1\times$; at $10\times$ that

only falls to 22%. The split is at 0%, and its error at 5 Lyapunov times is 1.18 — at the climatology level 1.00, i.e. saturating rather than exploding, exactly as the boundedness argument predicts. This is a guarantee from the function class, not a quantity that better optimisation can supply.

Which ingredient buys what. Model D holds boundedness fixed and removes the split. It reaches divergence 1.6% — essentially the split’s stability — but reconstruction $2.28\times$ floor and VPT 0.329, no better than the knob. So the two ingredients buy different things: **boundedness buys stability; separating the carriers buys the reconstruction and forecast quality.** Neither alone reproduces the result, which is the cleanest evidence we have that the split is doing real work rather than acting as a reparameterised ϕ .

Two-stage schedule. Training the identical modules end-to-end with the forecast objective included degrades reconstruction $4.4\times$ ($0.0026 \rightarrow 0.0117$) and VPT to 0.372. The architecture alone is not sufficient; the schedule that makes $\partial\mathcal{L}_B/\partial\theta_A = 0$ true is load-bearing.

Noise. Re-running the headline comparison at observation noise 0.15 and 0.30 (1 seed), the split leads the best ϕ at every level — gaps +0.163, +0.290, +0.208 in VPT — but the gap is *not monotonic*, peaking at 0.15. We report the sequence rather than a direction. Divergence for the split stays at 0% throughout.

Physics baseline. The PINN needed a fix to be a fair comparator: its decoder reads the raw latent, never the physically-calibrated coordinates, so nothing constrained that latent’s scale and a purely *relative* physics residual was satisfiable in a self-consistent but physically wrong coordinate system. Recalibrating the latent-to-state affine by least squares each epoch moved its forecast loss from 17.6 (rising during training) to 0.015 and its state-recovery R^2 from $[0.64, 0.70, 0.10]$ to $[0.996, 0.989, 0.981]$. Post-fix it is the strongest baseline on VPT (0.378) and has the best 50-Lyapunov-time attractor fidelity in the comparison (Wasserstein 0.068 vs the split’s 0.170).

5 Limitations

One system. All results are Lorenz-63 with one lift; the cross-domain phenomenon claim motivating this work is not tested here. **Compute-dependence is unresolved.** We did not run $10\times$ with 3 seeds, locate the crossover, or give the split $10\times$. **Fixed capacity.** The bounded carrier is a fixed 8×8 grid; the adaptive-precision mechanism motivating the broader hypothesis is untested. **Uneven seeds.** Single-seed rows report $\sigma = 0$ because $n = 1$, not because they are stable. **A floor artifact.** $\phi=0.25$ at $10\times$ sits at $0.99\times$ the measured floor; a capacity-matched control (latent=3) moves cleanly above it, indicating a wide unregularised latent exploiting cross-channel redundancy rather than a leak. **Constructed baselines.** The ϕ sweep and model D are the balancing recipe in its plainest form, not named published methods.

6 Conclusion

At matched parameters, wall clock and seeds, separating a bounded decoder-facing carrier from an unbounded forecaster-facing carrier reaches an operating point no setting of the weighted-loss knob reaches, and the knob’s own curve is nearly flat — within that family, sacrificing reconstruction buys almost no forecast. That is evidence the tradeoff, as practised, is an operating point rather than a frontier. But the advantage is compute-dependent, and we show it inverting at $10\times$ budget. The property that does not depend on compute is stability: bounded decoding gives 0% divergence by construction, where every unbounded baseline diverges on every rollout. We therefore read the honest conclusion as narrower than “the tradeoff dissolves”: it is not a single fixed frontier, compute-matched comparison alone cannot settle whether it is dissolvable, and the reliability half of the problem is separable from the accuracy half.

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172 A Full results

Table 2: Complete comparison, all 14 configurations. Epochs are seed-0 counts within the 120 s budget (Ours reports stage 1 + stage 2 combined).

Model	seeds	epochs	recon MSE	NRMSE	VPT ($\pm\sigma$)	diverge
$\phi=0.25$ (10 $\times$)	1	19	0.0025	0.0497	0.526	0.219
Ours (two-stage)	3	16	0.0026	0.0513	0.489 ± 0.013	0.000
PINN (true physics)	3	16	0.0047	0.0678	0.378 ± 0.019	0.260
$\phi=0.9$	1	2	0.0050	0.0706	0.335	1.000
AEGRU+sigmoid (D)	3	2	0.0057	0.0752	0.329 ± 0.004	0.016
$\phi=0.75$	1	2	0.0038	0.0618	0.326	1.000
$\phi=1$	1	2	0.0095	0.0971	0.326	1.000
$\phi=0.5$	3	2	0.0033	0.0572	0.326 ± 0.013	1.000
$\phi=0.1$	3	2	0.0027	0.0517	0.323 ± 0.011	1.000
$\phi=0.25$	3	2	0.0029	0.0535	0.320 ± 0.015	1.000
PINN ($\rho=26$)	1	16	0.0062	0.0788	0.245	0.000
persistence	—	—	—	—	0.036	0.000
$\phi=0$	1	2	0.0025	0.0501	0.027	1.000
climatology	—	—	—	—	0.000	0.000

173 Note the epochs column: the ϕ models complete ~ 2 epochs against Ours’ 16 in the same wall clock,
 174 because the joint model re-encodes full sequences every step while stage 1 trains on single timesteps
 175 and stage 2 on precomputed latents. Equal seconds is not equal gradient steps; this is the mechanism
 176 behind the budget result of §4, and it is a property of the architectures, not a protocol error.

177 B Long-horizon attractor statistics (50 Lyapunov times)

Model	Wasserstein $\downarrow$	spectrum log-err $\downarrow$	diverge	final NRMSE
PINN (true physics)	0.068	1.88	0.906	1.42
PINN ($\rho=26$)	0.149	2.07	0.188	1.42
Ours (two-stage)	0.170	2.06	0.000	1.44
AEGRU+sigmoid (D)	0.568	3.14	0.000	1.46
$\phi=1$	139.3	8.24	1.000	389.7
$\phi=0.5$	170.1	8.32	1.000	417.9
$\phi=0$	213.1	8.59	1.000	534.3

178 After 5,500 blind steps the split still traces the attractor (Wasserstein 0.170) while every weighted-loss
 179 setting is off by three orders of magnitude. The repaired PINN is best on this axis, as expected of a
 180 model handed the exact equations.

181 C Noise sweep

noise	model	$\times$ floor	VPT	diverge
0.05	Ours	1.06	0.489	0.000
0.05	$\phi=0.1 / \phi=0.5$	1.08 / 1.32	0.323 / 0.326	1.000
0.05	AEGRU+sigmoid (D)	2.28	0.329	0.016
0.15	Ours	0.93	0.598	0.000
0.15	$\phi=0.1 / \phi=0.5$	0.86 / 0.95	0.299 / 0.308	1.000
0.15	AEGRU+sigmoid (D)	1.08	0.290	0.000
0.30	Ours	0.92	0.462	0.000
0.30	$\phi=0.1 / \phi=0.5$	0.55 / 0.86	0.254 / 0.254	0.000 / 1.000
0.30	AEGRU+sigmoid (D)	0.92	0.245	0.000

182 At noise 0.15 and 0.30 several models fall below $1.0\times$ the measured floor. This is the same capacity
183 artifact discussed in the limitations, amplified: with more noise there is more to denoise, and cross-
184 channel redundancy lets a wide latent recover the clean signal slightly better than the per-channel
185 floor implies. It affects all models equally and does not change the ordering.

186 **D Reproducibility**

187 Code, data generator and the executed notebook (all outputs stored) accompany this submission. All
188 numbers come from a single end-to-end run at `QUICK=False` on one pinned CPU thread; the full
189 run takes approximately three hours from an empty checkpoint directory. Every model is cached
190 after training, so re-running reproduces the tables without retraining. Architecture: E is an MLP
191 $30 \rightarrow 128 \rightarrow 128 \rightarrow 30$; f is $30 \rightarrow 128 \rightarrow 64 \rightarrow 3$; m is $3 \rightarrow 64 \rightarrow 128 \rightarrow 64$ with a sigmoid; r is
192 $30 \rightarrow 128 \rightarrow 128 \rightarrow 64$ with a sigmoid; D is $64 \rightarrow 128 \rightarrow 128 \rightarrow 30$; g is a residual GRU ($3 \rightarrow 64$) with
193 a linear head. All activations GELU. Adam, $\eta = 10^{-3}$, $\lambda = 1$, stage-1 batch 256, stage-2 batch 64,
194 warm-up 64 steps, training rollout 20 steps, evaluation horizon 550 steps (5 Lyapunov times).